

Design Patent Prior Art Search via Citation Network Analysis and Multimodal LLM Pairwise Judgment

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Abstract

Prior art search for design patents requires identifying earlier designs that share salient visual features with a new application, but this task is difficult for conventional text-based retrieval because design patents typically contain little descriptive text beyond a brief claim. Although prior studies have developed image retrieval models for design patent search, it remains unclear whether recent multimodal large language models (MLLMs) can assess design similarity effectively without task-specific training. At the same time, citation network analysis has been widely used to study technological relationships among utility patents, yet design patent citation networks remain largely unexplored. This paper addresses these related gaps through an empirical study of the USPTO design patent corpus. We construct a citation network from examiner-cited patent pairs and analyze its structural properties, including registration year, classification code distribution, and single- versus multi-class assignments. We then evaluate whether an MLLM can judge the visual similarity of examiner-cited image pairs, use these judgments to refine noisy citation-derived ground truth, and compare the resulting positives with rankings obtained by encoding all patent images in a shared vector space. Specifically, we rank each image against the full corpus and evaluate retrieval under incomplete relevance judgments. Our results provide an initial empirical account of citation structure in design patents and suggest that zero-shot prompted MLLM-based visual similarity judgments can support interpretable prioritization of prior art candidates in design patent search.

1 Introduction

Design patents protect the ornamental appearance of products—including their shape, contour, and overall visual impression—rather than their functional features (Samuelson and Gergen, 2019). Unlike utility patents, which typically contain detailed claims and rich technical descriptions, design

patents often provide only a brief claim, commonly formulated as “the ornamental design for,” together with a limited set of figure captions (Shomee et al., 2024). Consequently, determining whether a new design is substantially similar to existing prior art depends primarily on visual comparison rather than textual analysis. This pronounced modality gap motivates computational approaches that can retrieve visually related designs and support interpretable prior art assessment.

A natural starting point for computational prior art search is to understand the structure of examiner-cited references: what characterizes citation counts in design patents and how citation relationships are distributed across design categories. Such structural characterization has been studied extensively for utility patent corpora, where network-analytic approaches—including exponential random graph models (ERGMs) and bibliometric citation analyses—have been used to examine knowledge flows, technological relationships, and citation mechanisms (Chakraborty et al., 2020). To the best of our knowledge, comparable structural analysis has not yet been applied to design-patent citation networks.

Existing computational approaches to design-patent retrieval have relied primarily on image or multimodal embedding models that require substantial domain adaptation. Kucer et al. (2022) introduced DeepPatent, a large-scale dataset for design-patent drawing recognition and retrieval, and showed that retrieval models benefit from training on domain-specific associations. Shomee et al. (2024) introduced IMPACT, a multimodal dataset with detailed captions, and reported that PatentCLIP—fine-tuned on design-patent image-caption pairs—outperforms general-purpose baselines on multimodal retrieval tasks. Lo et al. (2024) further proposed combining LLM-generated textual descriptions with distribution-aware contrastive learning to reduce dependence on man-

ual annotations, though this approach still requires a dedicated caption-generation and encoder-training pipeline. Taken together, these studies establish that domain adaptation improves retrieval performance, but at non-trivial data and training cost. A qualitatively different approach—presenting design-patent drawing pairs directly to a general-purpose MLLM with examiner-oriented prompts and asking it to assess ornamental similarity without task-specific fine-tuning—has not yet been systematically evaluated. A useful reference point is provided by Ikoma and Mitamura (2025), who showed that LLMs can assess novelty relationships from utility patent claim text; however, design patents call for visual comparison of ornamental features rather than analysis of claim language, and whether zero-shot prompted MLLM judgment can serve as an interpretable complement to embedding-based retrieval in this setting remains an open question.

2 Related Work

Large-scale design-patent datasets provide the empirical foundation for computational prior art search. Shomee et al. (2024) introduced IMPACT, a multimodal dataset that positions design-patent retrieval as a problem involving both patent drawings and associated textual information. Kucer et al. (2022) introduced DeepPatent for patent drawing recognition and retrieval at scale, highlighting the need for representations that capture visual rather than categorical similarity. These datasets make large-scale study of design patents possible, but they do not characterize how examiner-cited references are organized as a citation network, nor do they provide examiner-derived pairwise evidence of visual relatedness—both of which are central to the present study.

Citation network analysis offers a complementary perspective by treating prior-art relationships as relational data rather than isolated reference pairs. Network-analytic methods can characterize degree distributions, clustering, homophily, transitivity, and citation-formation mechanisms. Chakraborty et al. (2020) analyzed patent citation networks using descriptive statistics and ERGMs, demonstrating that network structure and patent attributes can be jointly modeled to reveal knowledge-flow patterns. This and related studies, however, focus primarily on utility patents or mixed patent corpora; design-patent citation networks,

whose edges are grounded in examiner-identified visual similarity rather than technical claim relationships, remain underexplored. The present study addresses this gap with network analysis tailored specifically to examiner-cited design-patent references.

On the retrieval side, existing approaches to design-patent prior art search rely mainly on learned visual or multimodal embeddings. Patent-CLIP, trained on IMPACT’s image–caption pairs, outperforms general-purpose vision–language baselines on multimodal retrieval tasks (Shomee et al., 2024). However, the IMPACT data are highly skewed across design-patent classes: a small number of classes contain many records, whereas numerous others remain comparatively sparse. This imbalance is important for retrieval evaluation because models may appear effective by capturing broad category-level regularities, even if they fail to distinguish fine-grained ornamental similarity within dense classes or to identify visually related designs across class boundaries. It is also important for citation analysis, since citation opportunities are unevenly distributed across the design-class system.

Lo et al. (2024) reduce dependence on manual annotations by combining LLM-generated image descriptions with distribution-aware contrastive learning, yet their framework still requires a dedicated captioning and encoder-training pipeline. A more direct alternative—prompting a general-purpose MLLM with examiner-oriented criteria to judge ornamental similarity from raw drawing pairs, without fine-tuning or intermediate captioning—has not been systematically explored. Ikoma and Mitamura (2025) demonstrate that LLMs can assess novelty relationships from utility patent claim text, but design patents require visual comparison of ornamental features rather than textual analysis of claims. The present study extends this line of work to zero-shot prompted visual judgment, using examiner-cited pairs as ground truth to evaluate how well MLLM assessments align with examination practice and how they compare with embedding-based retrieval rankings.

3 Method

Our empirical analysis is based on design-patent records from the IMPACT dataset (Shomee et al., 2024), which provides patent drawings and asso-

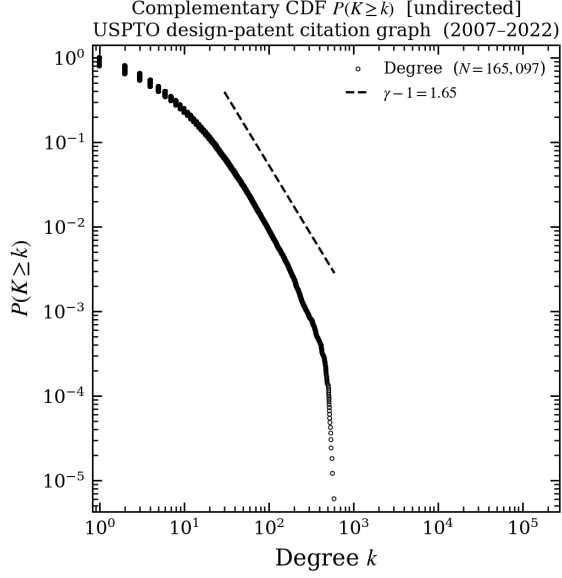


Figure 1: Complementary cumulative distribution function (CCDF) of the citation-network degree distribution.

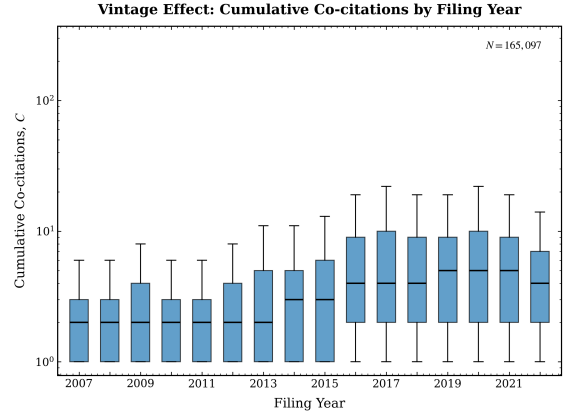


Figure 2: Cumulative co-citations C by filing year (log scale). Citation counts peak for mid-period cohorts and decline for recent years due to limited observation time.

ciated metadata for large-scale design-patent analysis. We link patent images, design-class labels, filing information, and examiner-cited references to create a unified corpus.

From the matched examiner-citation records, we construct a directed citation network in which each node represents a design patent and each edge points from a focal patent to a cited prior design. This representation converts examination records into a relational corpus and also defines citation pairs for subsequent similarity judgment.

Figure 1 reports the complementary cumulative distribution function of citation-network degree. This visualization emphasizes the tail of the distribution: a heavier right tail indicates that some designs are repeatedly connected through examiner-citation relationships, whereas the large mass of low-degree patents reflects the breadth of designs that rarely participate in such relationships.

We next examine how citation counts vary over filing-year cohorts Figure 2 and how examiner-cited relationships are distributed across design classes Figure 3, which shows the cross-class citation frequency for all examiner-cited pairs in the corpus. This strong class homophily suggests that design-patent examiners predominantly identify prior art within the same design category, providing an important baseline for understanding citation formation.

We next apply the MLLM to judge the visual similarity of each examiner-cited pair. Pairs judged

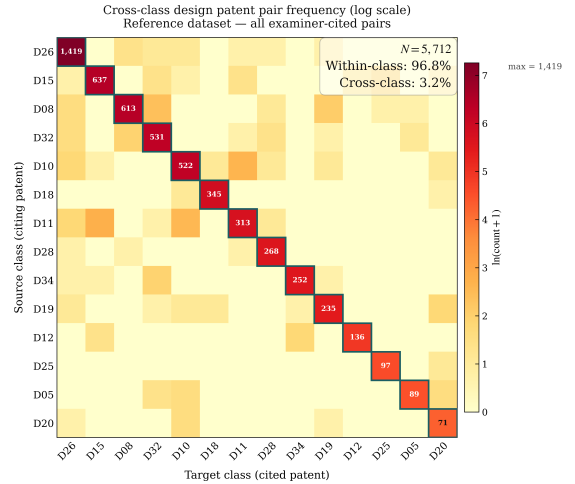


Figure 3: Cross-class citation frequency heatmap for the reference dataset of all examiner-cited pairs ($N = 5,712$, log scale). The dominant diagonal confirms that 96.8% of citations are within the same design class, with only 3.2% spanning class boundaries.

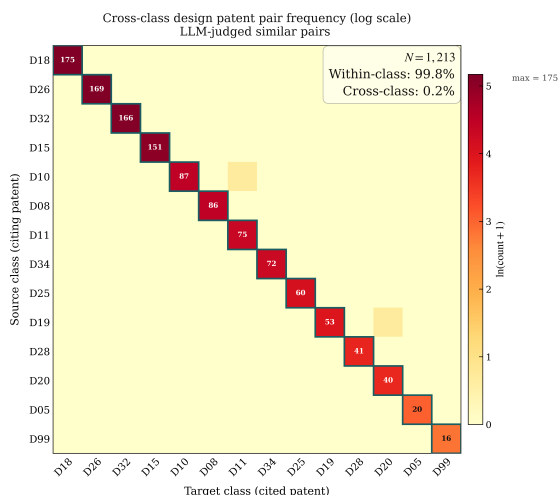


Figure 4: Cross-class citation frequency heatmap for LLM-judged similar pairs ($N = 1,213$, log scale). Within-class concentration rises to 99.8% (cross-class: 0.2%), indicating that MLLM visual similarity judgments select pairs that are more class-homogeneous than the full examiner-cited reference set.

as similar by the MLLM are retained as positive instances and used as ground truth for measuring retrieval rank in a vector-based search over the full design-patent image corpus. By analyzing the resulting rank distributions, we evaluate the quality of the MLLM-derived embeddings as a retrieval signal.

Figure 4 shows the corresponding heatmap for pairs judged visually similar by the MLLM. The within-class concentration increases further to 99.8%, with cross-class pairs reduced to 0.2%. This stronger class homogeneity indicates that MLLM visual similarity judgments filter toward pairs sharing a common design category more selectively than the full examiner-cited reference set, suggesting that zero-shot prompted visual judgment captures class-level ornamental coherence.

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A Example Appendix

This is an appendix.